Bhavesh Valecha

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RESEARCH INTERESTS

- Stochastic Thermodynamics
- Active Matter
- Nonequilibrium Statistical Physics
- Stochastic Processes

EDUCATION

Bachelor of Science & Master of Science

Indian Institue of Science Education and Research(IISER) Pune, India
August 2016 - July 2021
CGPA: **8.9/10**

RESEARCH PROJECTS

Thermodynamic Precision in Active Nematics

Supervisor: Prof. Dr. Klaus Kroy, Soft Condensed Matter Theory Group, Universität Leipzig
November 2022 - April 2023 | Leipzig, Germany

This project is aimed at, apart from developing other projects, studying the director field precision in the hydrodynamic description of active nematics. In particular, we want to investigate the effect of the striking feature of 'giant number fluctuations' in these systems on thermodynamic cost-precision trade-off relations like the TUR.

Violations of the Thermodynamic Uncertainty Relation (TUR)

Supervisors: Dr. Bijay K. Agarwalla, Assistant Professor, IISER Pune & Prof. Dr. Udo Seifert, Director, II. Institute for Theoretical Physics, University of Stuttgart
August 2020 - July 2021 | Pune, India

The master's thesis project focused on the violations of the TUR in open quantum systmes and the dependence of such bounds on system size. For a single quantum dot resonant transmission model, we showed that this is indeed the case. We also explored these bounds in the presence of a time-reversal symmetry breaking magnetic field.

Optimizing mesoscopic heat engines(DAAD-WISE Scholarship)

Supervisor: Prof. Dr. Udo Seifert, Director, II. Institute for Theoretical Physics, University of Stuttgart
May 2019 - July 2019 | Stuttgart, Germany

The project was aimed at studying the trade-off between the power output, efficiency and the fluctuations in power output using the trajectory approach of Stochastic Thermodynamics and Large Deviation Theory for an engine on the scale of Brownian particles. This trade-off was explored in more detail using the concept of the Pareto front that is commonly used in economics.

Reading on Nonequilibrium Statistical Physics

Supervisor: Prof. Deepak Dhar, Distinguished Visiting Faculty, IISER Pune
Spring 2019 | Pune, India

This reading project was aimed at gaining knowledge of the basic ideas and techniques in Nonequilibrium Statistical Physics and to acquire familiarity with associated tools and methods. In particular, the Exclusion Process and the paradoxical Brownian ratchet and pawl systems were studied in detail.

Effects of noise in the initial configuration of growing sandpiles

Supervisor: Prof. Deepak Dhar, Distinguished Visiting Faculty, IISER Pune

May 2017 - December 2018 — Pune, India

The primary aim was to study the effects of noise in the initial conditions of the centrally seeded Abelian Sandpile Model(ASM) on a square lattice. We studied patterns generated computationally using numerical methods and found that the average patterns were reminiscent of the well-studied problem of Rayleigh-Benard convection.

CONFERENCE PARTICIPATIONS

WE Heraeus' Quantum Thermodynamics for Young Scientists

02-06 Feb 2020 — Bad Honnef, Germany

COURSES TAKEN

PHYSICS

Theory: Classical & Relativistic Mechanics, Waves & Matter, Electrodynamics, Quantum Mechanics I & II, Mathematical Methods in Physics, Statistical Mechanics I & II, Condensed Matter Physics I & II, Quantum Field Theory, General Releativity, Atomic & Molecular Physics.
Extra-curricular: Stochastic Thermodynamics by Prof. Dr. Udo Seifert
Lab: Mechanics & Electromagnetism, Heat & Radiation, Optics, Modern Physics

MATHEMATICS

Single & Multi Variable Calculus, Linear Algebra, Introduction to Proofs, Probability & Statistics, Numerical Analysis, Special Structures in Mathematics, Graph Theory, Algorithms, Data Science

ACADEMIC ACHIEVEMENTS

Scholarships

DAAD-Working Intership in Science and Engineering (WISE) Scholar	**2019**
Kishore Vaigyanik Protsahan Yojana (KVPY) Fellow	**2017-2021**
DST INSPIRE Scholarship for Higher Education (SHE)	**2016-2017**
National Talent Search Examination (NTSE) Scholar	**2014-2016**

Talent Camps

VIJYOSHI National Science Camp	**2016**

Competitive Exams and Competitions

National Graduate Physics Examination 2017	**State Topper**
Kishore Vaigyanik Protsahan Yojana(KVPY) 2016	**All India Rank 35**
Joint Entrance Exam (Mains) 2016	**All India Rank 1773**
Maharastra Common Entrance Test	**Scored 189/200**

PROGRAMMING

C, Python, FORTRAN

EXTRA-CURRICULAR ACTIVITIES

Sports
Part of Football Team - Reached the semi-final of Intra IISER Football League 2018

Music
Can play the Ukulele

Community Service
Part of the 'Spread The Smile' Initiative 2017 for teaching kids in remote villages

Miscellaneous
- IISER Pune Quiz Club Coordinator (2017-18)
- Part of the Core Team of Physics question making body for a national level quiz conducted by IISER Pune called 'Mimamsa'

REFERENCES

- Prof. Dr. Udo Seifert — useifert@theo2.physik.uni-stuttgart.de
- Prof. Dr. Klaus Kroy — klaus.kroy@uni-leipzig.de